

VINOD OGUBOYINA

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PROFESSIONAL SUMMARY

Full Stack Cloud Engineer specializing in end-to-end web and mobile application development (React.js, React Native, Python/FastAPI), cloud deployment on AWS and Azure, and applied Gen AI/RAG systems. Comfortable owning a feature from API design and secure authentication through containerized deployment and production monitoring.

CORE TECHNICAL SKILLS

Backend & APIs :	Python, FastAPI, RESTful API design, JWT-based authentication & authorization, Microservices
Frontend Technologies :	React.js, HTML, CSS, JavaScript
Mobile Development :	React Native, Expo Go
Databases :	PostgreSQL, MongoDB, Redis
Cloud Platforms :	AWS (EC2, S3, RDS, Lambda, API Gateway, SES, CloudWatch), Azure (Functions, SQL Database, Monitor, App Insights, Blob Storage)
DevOps & Infrastructure :	Git, GitHub, Docker, Nginx (reverse proxy & load balancing), Kubernetes, CI/CD, Linux/Ubuntu Server Deployment, Shell scripting
AI Technologies :	RAG, LLM integration, Generative AI, LangChain, LangGraph, Conversational AI/Chatbots
Leadership :	Team Leadership, Requirements Gathering, System/Solution Architecture

PROFESSIONAL EXPERIENCE

Project Associate - CMTI (Central Manufacturing Technology Institute), Bengaluru

CMF Digitization Platform - Job-Shop Digitization Proposal (Team Lead)

- Led a cross-functional team to design a Job-Shop Digitization Platform replacing manual, paper-based shop-floor processes with an integrated digital workflow.
- Architected the solution around four core modules - Order Management (OMS), Product Data Management (PDM), Production Planning & Scheduling (PPS), and a tablet-based Work Execution Module (WEM).
- Defined the end-to-end workflow from Sales Order creation through automated multi-level BOM explosion, Work Order generation, capacity-based scheduling, and operator execution, ensuring full traceability from order to material.
- Drove requirements gathering with stakeholders to document current-state (AS-IS) gaps and define the functional specification and event-driven architecture for the proposed (TO-BE) system.
- Developed a shop-floor Q&A chatbot using LangChain and LangGraph that retrieves and returns existing, pre-validated answers to operator queries, reducing dependency on supervisors for repetitive shop-floor questions.
- Handled server-side deployment of the CMF software on an Ubuntu/Linux server, managing environment setup, configuration, and hosting for shop-floor use.

BEL MES - Manufacturing Execution System

- Developed the Document Management module end-to-end with React.js frontend and Python FastAPI backend, including versioning and role-based access control secured with JWT authentication.
- Built the Inventory Management module for tracking materials, stock levels, and warehouse operations, backed by RESTful APIs for inventory transactions.
- Deployed the application on Azure with PostgreSQL for persistence; configured Nginx as a reverse proxy in front of the app servers.
- Set up centralized logging and monitoring using Azure Monitor and Log Analytics for proactive troubleshooting.

Cloud-Based Logistics Management System

- Built a full stack application (React.js + FastAPI) for material movement requests with real-time tracking via AWS Lambda and API Gateway.
- Deployed on AWS (EC2, S3, RDS/PostgreSQL) behind Nginx, containerized with Docker and orchestrated with Kubernetes.
- Configured a Git-based CI/CD pipeline for automated build, test, and deployment on every push to GitHub.
- Implemented JWT-secured RESTful APIs for reliable frontend-backend communication and data management.

AI-Powered Tool Management Platform

- Built a full stack solution (React.js + FastAPI) with a RAG-based Gen AI system for intelligent tool-request processing and recommendations.
- Integrated an LLM into automated approval workflows to support faster, more consistent decision-making.
- Deployed on Azure using Azure Functions and Azure SQL Database; instrumented with Azure Monitor and Application Insights.

Automated Calibration Management System

- Developed a full stack web app (HTML/CSS/JavaScript frontend, Python backend) for calibration scheduling and tracking, exposed via RESTful APIs.
- Deployed on AWS with MySQL and automated email notifications using SES and Lambda; used CloudWatch for performance monitoring and alerting.

AI Inspection Data Processing Platform

- Built an end-to-end data processing pipeline (React.js + FastAPI) containerized with Docker and backed by MongoDB for flexible storage.
- Implemented data storage across Azure Blob Storage and AWS S3, and an automated reporting system with LLM-generated insights.

EDUCATION

B.Tech, Computer Science and Engineering - 2022 - 2025

NBKR Institute of Science and Technology, Andhra Pradesh - CGPA: 7.96

Diploma in Civil Engineering - 2019 - 2022

CBIT (State Board of Technical Education, AP) - 66.57%

CERTIFICATIONS

- AWS Certified Solutions Architect
- Microsoft Azure Fundamentals
- Python Programming - Infosys Springboard (2024)

LANGUAGES

English - Professional Working Proficiency | Telugu - Native Speaker | Kannada - Fluent